State-Level Consumer Spending Analysis — Module Summary

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1. Overview

This report documents a reproducible Python statistical analysis examining state-level daily consumer spending across five Northeastern US states — Connecticut, Massachusetts, New Jersey, New York, and Rhode Island — between January 2020 and June 2024. The analytical objective was to determine whether mean daily retail spending (including grocery) differs significantly across state boundaries. The dataset was sourced from the Data.gov catalog entry *Percent Change in Consumer Spending* (State of Connecticut, https://catalog.data.gov/dataset/percent-change-in-consumer-spending), which publishes aggregated and anonymized card-transaction data compiled by Opportunity Insights from Affinity Solutions. The full workflow was implemented in `analysis.ipynb` following the Initial Data Analysis framework proposed by Huebner et al. (2018), which emphasizes structured data validation and assumption-checking prior to inferential testing.

2. Dataset Description

The dataset contains 4,905 daily observation records distributed evenly across five states (981 records per state, 20% each), with 12 columns capturing merchant-level spending indices. The key variables examined were `retail_inc_grocery` (total retail spending including grocery), `grocery_spending`, `food_service_spending`, `transportation_spending`, and additional merchant categories such as `recreation_spending` and `healthcare_spending`. A `state_name` categorical grouping variable was derived from FIPS codes (9 = CT, 25 = MA, 34 = NJ, 36 = NY, 44 = RI) to enable group-level comparisons. All spending values are expressed as relative indices benchmarked against a baseline period rather than absolute dollar volumes. Data validation confirmed zero missing values across all 12 columns.

3. Methods

All computations were performed in Python using Pandas (McKinney, 2010), NumPy (Harris et al., 2020), Matplotlib (Hunter, 2007), and Seaborn (Waskom, 2021), with inferential testing via SciPy.

Descriptive statistics. Central tendency and dispersion were computed using grouped aggregations (`mean`, `std`, `median`, and interquartile range) rather than a single global `describe()` call. Grouped summaries were preferred because the analytical question concerns *between-state* differences; global statistics would obscure group-level variation — a data-quality principle emphasized by Huebner et al. (2018).

Visual models. Three complementary visualizations were constructed: (1) a box plot of retail spending by state to reveal median shifts and dispersion; (2) a correlation heatmap to assess inter-category co-movement; and (3) a scatter plot of grocery versus food-service spending colored by state. The box plot was designated the primary visual model because it directly mirrors the group structure evaluated in the hypothesis test.

Hypothesis test. A one-way Analysis of Variance (ANOVA) was selected because the design compares the means of one continuous dependent variable (`retail_inc_grocery`) across five independent categorical groups (`state_name`), with each observation belonging to exactly one group (Kim, 2017).

- **H_0:** μ_CT = μ_MA = μ_NJ = μ_NY = μ_RI
- **H_1:** At least one state population mean differs from the others

Assumptions checked. Sample independence was satisfied by the panel structure; normality was supported by the Central Limit Theorem given n = 981 per group. Homogeneity of variance was tested using Levene's Test (Huebner et al., 2018), which failed ($W = 33.51$, $p = 1.28 \times 10^{-27}$), driven by Rhode Island's elevated dispersion. Consequently, Welch's ANOVA was conducted as a heteroscedasticity-robust alternative (Delacre et al., 2019). Post-hoc pairwise comparisons were performed using Tukey's HSD to identify which specific state pairs differed.

4. Results

Descriptive findings. Massachusetts (mean = 22.21) and New Jersey (mean = 21.17) recorded the highest mean retail spending indices, while New York (18.90) and Rhode Island (17.95) recorded the lowest. Rhode Island also exhibited the greatest dispersion (SD = 14.38, IQR = 18.23), whereas New York was the most stable (SD = 9.74).

Hypothesis test results. Both models produced consistent, highly significant decisions:

Test	F-statistic	p-value	Decision
Standard One-Way ANOVA	$F(4, 4900) = 22.49$	1.99×10^{-18}	Reject H_0
Welch's ANOVA	$F(4, 2441.13) = 23.11$	9.09×10^{-19}	Reject H_0

Because both p-values fall far below $\alpha = 0.05$, the null hypothesis is rejected: state location is a statistically significant driver of mean retail spending.

Post-hoc Tukey HSD. Pairwise tests revealed a three-tier structure: Massachusetts and New Jersey formed a statistically indistinguishable high tier ($p = 0.264$); New York and Rhode Island formed a statistically indistinguishable low tier ($p = 0.348$); and Connecticut occupied an intermediate position, differing significantly from MA ($p < 0.001$) and NJ ($p = 0.004$) but not from NY ($p = 0.907$) or RI ($p = 0.052$).

5. Interpretation for a Non-technical Audience

The analysis set out to answer a simple question: do people in different Northeastern states spend noticeably different amounts on everyday retail purchases? The answer is **yes, and the difference is not a matter of luck**. Massachusetts and New Jersey consistently showed the highest levels of retail spending — about 22% above the 2020 baseline. New York and Rhode Island trailed behind by roughly four percentage points. Connecticut sat comfortably in the middle.

The statistical tests confirm that these gaps are real. The chance that the observed differences arose purely from random daily fluctuation is less than one in a quadrillion — an essentially impossible coincidence. However, spending also behaves differently in different places. Rhode Island's spending was the most erratic, swinging widely from day to day, whereas New York's was the steadiest.

In short: **consumers in neighboring states do not behave identically.** An economic policy or marketing strategy designed for Massachusetts consumers may not transfer directly to Rhode Island consumers, even though the two states share a border.

6. Limitations and Potential Bias

Geographic constraint. All observations are drawn from five Northeastern states characterized by high population density and elevated cost of living. Findings may not generalize to the South, Midwest, or rural economies (Huebner et al., 2018).

Card-transaction proxy bias. Spending metrics are derived from aggregated credit and debit card feeds. Cash-dominant transactions, informal economic activity, and unbanked or underbanked demographic segments are systematically underrepresented — introducing a selection bias into the measurement itself (Delacre et al., 2019).

Relative indexing. All values are index points relative to a baseline, not absolute dollar volumes. Direct monetary comparison between merchant categories is therefore not possible, and absolute revenue magnitudes cannot be reconstructed from this data.

Ecological fallacy. State-level aggregation conceals significant intra-state variation between metropolitan centers (e.g., New York City, Boston) and rural counties. Conclusions drawn at the state level should not be applied to individual consumers or local markets — a bias discussed extensively in the reproducibility and inference literature (Pineau et al., 2021).

7. Reproducibility

To ensure complete scientific reproducibility, the analysis environment was isolated in a dedicated Python virtual environment and dependencies were pinned in **requirements.txt** using **pip freeze**. The pinned file captures the exact versions of the core analytical libraries (**pandas, numpy, scipy, matplotlib, seaborn, and jupyter/notebook**) alongside all transitive dependencies, ensuring the notebook executes identically on any machine where the environment is recreated via pip install -r requirements.txt.

All dataset parsing, column standardization, and state-name mapping logic are implemented as explicit, deterministic transformations within the notebook itself — no manual data edits, no hidden preprocessing outside the notebook, and no hard-coded intermediate results. This design follows the reproducibility guidance of Huebner et al. (2018), which recommends that all data cleaning and derivation steps be made explicit and auditable, and aligns with broader best practices for reproducible data science workflows (Pineau et al., 2021). The inclusion of **requirements.txt** and the original CSV dataset alongside the notebook ensures that any reviewer can re-execute the full analysis top-to-bottom and arrive at identical descriptive statistics, figures, and hypothesis-test results.

8. Summary and Interpretation

In summary, exploratory and inferential analysis of the Opportunity Insights consumer spending dataset reveals that state-level location is a statistically significant driver of retail spending behavior across the five Northeastern states examined (N = 4,905 daily observations; 2020–2024). Massachusetts and New Jersey maintained the highest mean retail spending indices (approximately 22.21 and 21.17 respectively), while New York and Rhode Island recorded the lowest (approximately 18.90 and 17.95). Connecticut occupied an intermediate tier. These differences were confirmed by both a standard one-way ANOVA ($p = 1.99 \times 10^{-18}$) and a heteroscedasticity-robust Welch's ANOVA ($p = 9.09 \times 10^{-19}$), with post-hoc Tukey HSD tests confirming a three-tier structure in which MA and NJ form a statistically indistinguishable high group and NY and RI form a statistically indistinguishable low group.

The analytical decisions underpinning these findings — grouped descriptive statistics, three complementary visual models, explicit assumption checking via Levene's Test, and dual-ANOVA robustness testing — were designed to support reproducibility and interpretability in equal measure. At the same time, practical application of these insights must account for key responsible practice considerations, notably the geographic restriction to the Northeastern US, the credit/debit-card proxy bias that underrepresents cash and unbanked transactions, the relative index (rather than dollar-volume) nature of the spending metric, and the ecological fallacy risk inherent in state-level aggregation.

9. Future Integration and Reflections

Future extensions of this analysis should focus on three directions: predictive modeling, richer segmentation, and automation.

Predictive modeling. A natural next step would be to construct a state-level or day-level forecasting pipeline using time-series models (e.g., SARIMAX or gradient-boosted regressors) to predict retail spending index values from lagged spending, calendar features, and macroeconomic indicators. Because the class distribution of states is balanced by design (981 records each), a simple stratified train-test split on time would preserve representation; standardization of the spending indices via StandardScaler should precede any distance-based or regularized model.

Segmentation. Clustering the five states on their full merchant-category profiles would allow data-driven verification of the three-tier structure revealed by Tukey HSD, and could be extended to include county- or MSA-level data to address the ecological fallacy limitation directly.

Automation and agentic workflows. Agentic automation offers clear value for automating routine exploratory data analysis (EDA), hyperparameter sweeps, and anomaly detection in the daily series (e.g., flagging pandemic-period discontinuities). However, autonomous re-derivation of spending categories, unsupervised feature selection on the relatively small Rhode Island series, or auto-generated economic interpretations without geographic caveats would pose severe analytical risks. Any

autonomous component should therefore be human-supervised at the final interpretation stage — consistent with the reproducibility and oversight principles advocated by Pineau et al. (2021).

10. Figure Interpretations

Grounded statistical aggregations and graphical interpretations demonstrate significant geographic differentiation across state-level retail spending. The grouped summary by state is as follows: Massachusetts recorded a mean retail index of 22.21 with standard deviation 10.19, median 23.90, and interquartile range 11.00. New Jersey recorded a mean of 21.17 with standard deviation 10.82, median 23.80, and interquartile range 15.00. Connecticut recorded a mean of 19.35 with standard deviation 11.60, median 21.10, and interquartile range 14.80. New York recorded a mean of 18.90 with standard deviation 9.74, median 20.70, and interquartile range 12.40. Rhode Island recorded a mean of 17.95 with standard deviation 14.38, median 17.90, and interquartile range 18.23.

Figure 1 (Box Plot — Retail Spending Distribution by State). The box plot reveals distinct regional variation in retail spending. Massachusetts (median = 23.90) and New Jersey (median = 23.80) display the highest spending baselines, while Rhode Island (median = 17.90) exhibits the lowest median but the greatest dispersion (IQR = 18.23, SD = 14.38). New York (SD = 9.74) is the most tightly clustered distribution. The visual directly mirrors the group structure tested by the ANOVA, making it the primary visual model supporting the inferential conclusion.

Figure 2 (Correlation Heatmap — Inter-Category Relationships). Apparel and general merchandise spending exhibits the strongest linear coupling with total retail spending (r = +0.82), confirming that clothing and merchandise purchases dominate the retail index. Food service (r = +0.58), grocery (r = +0.56), and healthcare (r = +0.55) show moderate positive co-movement, while transportation (r = +0.32) exhibits a comparatively weak relationship, suggesting transportation spending is driven by distinct factors such as commuting and travel rather than retail shopping activity.

Figure 3 (Scatter Plot — Food Service vs. Grocery by State). State clustering is clearly visible. Connecticut dots concentrate at higher grocery spending (mean = 23.53), whereas Rhode Island dots cluster in the lower-left quadrant, reflecting the state's low grocery (mean = 12.88) and pronouncedly negative food service (mean = −17.41) spending indices. This confirms that state-level spending behavior is not interchangeable — geography materially shapes consumer transaction patterns.

11. References

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